

AIM ASEAN Programme: AI for MSME Advancement in ASEAN

Curriculum Outline

Curriculum Designed and Developed by:
AIM ASEAN Programme

Confidentiality Notice:

This curriculum is specifically designed for the AIM ASEAN Programme and is intended solely for the training and development of MSMEs in the ASEAN region. Unauthorized use or reproduction of this material is prohibited.

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Table of Contents

Curriculum Outline for AIM ASEAN Programme	3
Guidance on Duration and Implementation	3
Our Design Principles	4
Essential Modules	5
Module 1: Artificial Intelligence (AI) - The Basics and Use Cases	5
Module 2: The AI Adoption Journey - Steps for Business Owners	7
Module 3: Using AI Responsibly - Ethical & Governance Principles	10
Module 4: Key Considerations & Framework for Successful AI Deployment	13
Optional Modules	15
Module 5: Advanced Data Management & Governance for AI	15
Module 6: Building an AI-Ready Culture & Workforce Transformation	17
Module 7: AI Project Risk Management & Mitigation	19
Acknowledgement	21

Curriculum Outline for AIM ASEAN Programme

AI for MSME Advancement in ASEAN, or AIM ASEAN, is a regional programme that helps Micro, Small, and Medium Enterprises (MSMEs) in Southeast Asia use artificial intelligence (AI) to grow stronger, more efficient, and more competitive. Led by the ASEAN Foundation in partnership with AVPN and supported by [Google.org](https://www.google.org) and ADB, the two-year programme will provide AI training to MSMEs across all ten ASEAN Member States.

This curriculum aims to demystify AI, guide business owners through the adoption journey, and ensure that they not only understand AI's potential but also implement it effectively and ethically. Participants will be equipped with the skills to identify AI opportunities in their business, make informed decisions about AI adoption, and ensure they are deploying AI responsibly. By the end of this programme, MSMEs will be empowered to scale their operations with AI-driven strategies and gain a competitive edge in the fast-evolving digital landscape.

The following detailed modules will take participants step-by-step through the various stages of AI integration, from understanding AI basics and business applications, to navigating ethical considerations, ensuring data governance, and preparing their workforce for the future of AI. This structured approach ensures that business owners can implement AI seamlessly into their daily operations, maximise its potential, and drive long-term success.

Guidance on Duration and Implementation

Each module within this curriculum includes suggested durations for the subtopics, providing a guideline for the ideal time allocation based on the content's complexity and learning objectives. These time frames are intended to assist AIM ASEAN Local Implementing Partners (Local implementers) in structuring the training sessions effectively and ensuring that key concepts are covered thoroughly. However, these durations are flexible and should not be viewed as rigid constraints. Local implementers are encouraged to adapt the timing as needed to best suit the learning pace, participant engagement, and regional context. The goal is to ensure that the curriculum remains relevant and impactful, while offering flexibility for customization in different ASEAN member states. Additionally, implementers may refer to the [Resources Hub](#) for supplementary content, materials, and tools to support their delivery and enhance the learning experience for participants.

The curriculum consists of four essential modules and three optional modules designed for flexibility in local implementation. The essential modules are required and must be covered by all implementers, as they provide foundational knowledge and guidance critical for AI adoption. These include:

- Module 1: Artificial Intelligence (AI) – The Basics and Use Cases,
- Module 2: The AI Adoption Journey – Steps for Business Owners,
- Module 3: Using AI Responsibly – Ethical & Governance Principles, and
- Module 4: Key Considerations & Framework for Successful AI Deployment.

The optional modules are designed for enterprises whose characteristics, such as company size or AI literacy level, make them suitable for deeper exploration of these topics. Local implementers may select these optional modules based on the needs and readiness of participating enterprises:

- Module 5: Advanced Data Management & Governance for AI,
- Module 6: Building an AI-Ready Culture & Workforce Transformation, and
- Module 7: AI Project Risk Management & Mitigation.

Our Design Principles

The following design principles guide the AIM ASEAN curriculum, ensuring it is practical, relevant, and actionable for MSMEs on their AI adoption journey.

1. **Encourage AI Adoption by Combining Literacy with Business Relevance:** After getting all the MSMEs on the same page with a quick revision on foundational AI, we will highlight tangible benefits and strategic implications, helping participants see AI as a vital tool for their company's growth rather than just a complex technology. The focus is on tackling and motivating MSMEs to adopt AI into their business processes.
2. **Showcase Practical and Impactful Use Cases:** We will utilize a diverse range of real-world examples and relevant use cases from various business sectors, such as agriculture, manufacturing, and food & beverage. These examples will illustrate how AI can be directly applied to enhance operations, marketing, customer service, and overall productivity, demonstrating clear and actionable pathways for adoption.
3. **Provide Clear, Concise Steps for Implementation:** The curriculum will guide participants through the AI adoption journey with clear and concise steps. We will focus on a practical, hands-on approach that enables them to adopt AI tools and launch small-scale projects without feeling overwhelmed, ensuring they leave with an actionable plan to begin their AI journey. This principle will serve as the foundation for developing the program's practical material.

Essential Modules

Module 1: Artificial Intelligence (AI) - The Basics and Use Cases

Suggested Duration: 77 mins

This module will do a quick revision on what Artificial Intelligence is so far. We'll explore the different types of AI, such as Machine Learning, Deep Learning, and Generative AI, with practical examples for each. The module also covers the key business reasons for adopting AI, like increasing efficiency, improving customer experience, and gaining a competitive edge. Finally, we'll dive into common AI use cases across different business functions, including customer service, marketing, operations, and finance.

Goal: Provide the foundational knowledge for Business Owners to be aware of AI and its impact on businesses. This encourages MSMEs to start looking to adopt AI into their business.

Learning Objectives:

Upon completion of this module, business owners will be able to:

- Understanding what AI is about.
- Distinguish between the main types of AI (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI) and recognize common examples of each.
- Articulate at least five compelling business reasons for considering AI adoption (e.g., efficiency, improved decision-making, enhanced customer experience).
- Identify and give examples of how AI can be applied to common business functions across various departments (e.g., customer service, marketing, operations, finance).
- Evaluate potential areas within their own business where AI could generate significant value and kick-start their AI adoption.

Module Outline:

- **1.1 What is AI?**
 - **1.1.1 Revising the Basics (8 mins):** What is underneath AI? What does it need to work? Good quality data, Selection of algorithms, and structuring the output.

- **1.2 Types of AI (Simplified)**
 - **1.2.1 Machine Learning (ML) (8 mins):** AI that learns from data without explicit programming. Supervised (labelled data), Unsupervised (unlabelled data), Reinforcement (behavior)
 - *Examples:* Recommendation engines (e.g., Netflix), fraud detection.
 - **1.2.2 Deep Learning (DL) (5 mins):** A subset of ML, using neural networks for complex pattern recognition. The general structure of a neural network, including input, hidden, and output layers, followed by Deep Learning.
 - *Examples:* Image recognition, natural language processing.
 - **1.2.3 Natural Language Processing (NLP) (3 mins):** AI that understands, interprets, and generates human language. Two areas - Natural Language Generation & Natural Language Understanding
 - *Examples:* Chatbots, sentiment analysis, language translation.
 - **1.2.4 Computer Vision (CV) (3 mins):** AI that enables computers to "see" and interpret visual information. Example Yolo V3.
 - *Examples:* Facial recognition, quality control in manufacturing.
 - **1.2.5 Generative AI (GenAI) (5 mins):** AI that creates new content (text, images, audio, code).
 - *Examples:* Content creation, design, personalized marketing messages.
- **1.3 Why Adopt AI? The Business Imperatives -**
 - **1.3.1 Increased Efficiency & Automation (3 mins):** Automating repetitive tasks, freeing up human resources.
 - **1.3.2 Enhanced Decision-Making (3 mins):** Data-driven insights for better strategic and operational choices.
 - **1.3.3 Improved Customer Experience (3 mins):** Personalization, 24/7 support, faster response times.
 - **1.3.4 Innovation & New Revenue Streams (3 mins):** Creating new products/services or optimizing existing ones.
 - **1.3.5 Competitive Advantage (3 mins):** Staying ahead in a rapidly evolving market.
- **1.4 Common Business Use Cases for AI**
 - **1.4.1 Services (5 mins):** Chatbots, virtual assistants, automated support.
 - **1.4.2 Marketing & Sales (5 mins):** Personalized recommendations, lead generation, sentiment analysis.
 - **1.4.3 Operations & Supply Chain (5 mins):** Predictive maintenance, inventory optimization, logistics.
 - **1.4.4 Finance & Accounting (5 mins):** Fraud detection, risk assessment, automated reconciliation.

- **1.4.5 Human Resources (5 mins):** Automated screening, personalized learning paths.
- **1.5 Exercise: Gemini and NotebookLM (10 mins):** Hands-on practice using Google's Gemini and NotebookLM to research a topic, extract key insights, and turn the findings into well-structured content.

Module 2: The AI Adoption Journey - Steps for Business Owners

Suggested Duration: 79 mins

This module provides a practical, step-by-step roadmap for business owners looking to implement AI. You'll learn to define clear business goals and identify opportunities where AI can make the biggest impact. The module guides you through assessing your company's data readiness, exploring different solution options (like "buy vs. build"), and developing a small pilot project to test the AI's value. The module concludes with a focus on scaling up your AI project and integrating it seamlessly with your existing systems and team.

Goal: Provide a practical, step-by-step roadmap for implementing AI in a business.

Learning Objectives:

Upon completion of this module, business owners will be able to:

- Outline a structured, five-step process for initiating and deploying an AI project in their business.
- Explain the importance of defining clear business goals before exploring AI solutions.
- Assess their organization's data readiness for AI initiatives.
- Evaluate different AI solution acquisition strategies (build, buy, AlaaS, partner) based on business needs.
- Understand the value of starting with pilot projects and the importance of continuous monitoring.

Module Outline:

- **2.1 Step 1: Define Clear Business Goals & Identify Opportunities**
 - **2.1.1 Problem-First Approach (5 mins):** Don't adopt AI for AI's sake. What specific problems can AI solve? Ideally, should be using AI to improve a measurable outcome. Give some examples.
 - **2.1.2 Prioritize High-Impact Areas (5 mins):** Start with small, manageable projects that can deliver measurable ROI. Ideally, the use cases should be revenue-generating rather than, cost savings or improving the intangibles.
 - **2.1.3 Set SMART Goals (4 mins):** Specific, Measurable, Achievable, Relevant, Time-bound. If it cannot be measured, it cannot be improved. Need good quality data to calculate the performance metrics.
- **2.2 Step 2: Assess Data Readiness**
 - **2.2.1 Data is Fuel (4 mins):** AI needs data. Understand what data you have, where it is, and its quality.
 - **2.2.2 Data Collection & Storage (5 mins):** Identify necessary data, ensure ethical collection, and secure storage. There is a need to have data that can answer business questions. Having good quality data starts from designing the collection methods.
 - **2.2.3 Data Quality & Governance (7 mins):** Importance of clean, accurate, and consistent data. What are the different dimensions to measure data quality, besides accuracy? The benefits of having good quality data, and why the need for data governance.
- **2.3 Step 3: Explore AI Solutions - Build vs. Buy vs. Partner**
 - **2.3.1 Off-the-Shelf Solutions (4 mins):** Ready-made AI tools (e.g., CRM with AI features, marketing automation platforms). What are the advantages and disadvantages of buying off-the-shelf?
 - **2.3.2 Custom Development (4 mins):** Building AI solutions tailored to specific needs (often requires external expertise). What are the advantages and disadvantages of doing custom development?
 - **2.3.3 General Considerations (6 mins):** Software, Hardware, Skills & Knowledge to unlock and maintain the value from AI Solutions.
 - **2.3.4 AI-as-a-Service (AlaaS) (6 mins):** Cloud-based AI services from providers (e.g., Google Cloud AI, AWS AI). Showcase a few of these providers. What are the advantages and disadvantages of using Cloud-Based AI?
 - **2.3.5 Partnering with Experts (4 mins):** When to engage AI consultants or development firms. The benefits and risks of hiring consultants. How to select the consultants.

- **2.4 Step 4: Develop a Pilot Project & Prototype**
 - **2.4.1 Start Small (3 mins):** Test AI in a controlled environment to validate its value and learn.
 - **2.4.2 Iterative Approach (3 mins):** Implement, test, gather feedback, refine, and repeat. Any projects that involves data and AI will need experimentation.
 - **2.4.3 Measure Performance (5 mins):** Track KPIs to assess the pilot's success. Going forward, the POC's performance will be used as a benchmark.
- **2.5 Step 5: Scale & Integrate**
 - **2.5.1 Integrate with Existing Systems (5 mins):** Ensuring seamless flow of data and operations. Planning is still needed to reduce the disruption to BAU processes.
 - **2.5.2 Employee Training & Change Management (5 mins):** Educating and engaging staff, addressing concerns, and fostering adoption. Start with managing staff uncertainty of the future, and how the tools implemented will be helping them. Promote an open culture of feedback, that the feedback is used solely to improve the adoption of AI.
 - **2.5.3 Continuous Monitoring & Improvement (4 mins):** AI models need ongoing maintenance and refinement. Briefly talk about model drift, and also that GenAI models are improved with new data and training methods.

Module 3: Using AI Responsibly - Ethical & Governance Principles

Suggested Duration: 79 mins

This module is designed to help you understand the important ethical considerations of using AI. It introduces core principles like fairness, transparency, and accountability, with clear examples to illustrate what each one means in a business context. You'll learn how to identify and mitigate common risks, such as algorithmic bias and data quality issues. The module also provides a simplified framework for setting up AI governance, helping you establish clear policies and responsibilities for using AI in your organization.

Goal: Understand the ethical implications of AI and establish principles for responsible deployment.

Learning Objectives:

Upon completion of this module, business owners will be able to:

- Identify and explain the five core principles of responsible AI (fairness, transparency, accountability, privacy, human oversight).
- Recognize common ethical risks associated with AI (e.g., algorithmic bias, "black box" problems, misuse).
- Outline basic steps for establishing a simplified AI governance framework within their small to medium-sized business, including policy definition and oversight.
- Understand the importance of continuous monitoring for ethical adherence and risk mitigation.

Module Outline:

- **3.1 Core Principles of Responsible AI**
 - **3.1.1 Fairness & Non-Discrimination (6 mins):** Ensuring AI systems do not perpetuate or amplify existing biases. Share some examples, such as Amazon's gender bias hiring, COMPAS bias.
 - *Example:* Checking hiring AI for gender/racial bias.
 - **3.1.2 Transparency & Explainability (XAI) (6 mins):** Understanding how AI makes decisions. Explain why this is challenging if models like Support Vector Machines or Deep Learning models are used.
 - *Example:* Being able to explain why a loan application was approved or denied by AI.
 - **3.1.3 Accountability (4 mins):** Clearly defining who is responsible for AI outcomes and potential errors. State why this could be challenging, perhaps with an example of a self-driving car.

- *Example:* Establishing a protocol for human review of critical AI decisions.
 - **3.1.4 Privacy & Security (5 mins):** Protecting sensitive data used by and generated from AI. State the recent case of ChatGPT conversation threads can be found on Google Search.
 - *Example:* Ensuring compliance with data protection regulations (e.g., GDPR, PDPA in Singapore).
 - **3.1.5 Human Oversight & Control (5 mins):** Ensuring humans remain in the loop for critical decisions and can override AI. State some guidelines on when there should be a human-in-the-loop, for instance, medical care, education, etc.
 - *Example:* Not fully automating customer complaints without a human escalation path.
- **3.2 Identifying and Mitigating AI Risks**
 - **3.2.1 Algorithmic Bias (5 mins):** How biases can enter AI systems through data or design. Example: Amazon gender bias hiring.
 - *Action:* Regular auditing of data and model outputs.
 - **3.2.2 Data Quality & Integrity (5 mins):** The impact of poor data on AI performance and ethics. The need to measure AI performance to detect the bias as early as possible.
 - *Action:* Implementing robust data governance and cleansing processes.
 - **3.2.3 "Black Box" Problem (5 mins):** Challenges in understanding complex AI models. State examples of easy-to-explain models like Decision Trees, and complex models like Support Vector Machine and Deep Learning
 - *Action:* Prioritizing explainable AI tools where transparency is critical.
 - **3.2.4 Misuse & Malicious Use (5 mins):** Preventing AI from being used for harmful purposes.
 - *Action:* Establishing clear use policies and monitoring.
 - **3.2.5 Environmental Impact (6 mins):** Considering the energy consumption of large AI models. Share news reports on energy and water consumption of data centers.
 - *Action:* Opting for energy-efficient solutions where possible.
- **3.3 Building an AI Governance Framework (Simplified for MSMEs)**
 - **3.3.1 Define AI Use Policies (5 mins):** Clear guidelines for employees on acceptable AI tools and data usage. Guidelines should be tied back to the mission and values of the business, define why you are setting up the policies e.g. ensure that our customer's benefits come first. Where possible,

previous use cases should be detailed and documented to be kept as a knowledge base, to inform on the AI use policies design.

- **3.3.2 Assign Ownership (4 mins):** Designate individuals or teams responsible for AI strategy, risk, and compliance. Use a framework like RACI (Responsible, Accountable, Consulted, Informed) to clearly map out who is responsible for each part of the AI project lifecycle. With ownership comes accountability and action. Ownership needs to be clear with the roles and responsibilities that come with the ownership title.
- **3.3.3 Implement Review Processes (4 mins):** Regular checks for ethical adherence, performance, and compliance. Consider setting up an AI Governance committee. The frequency of meetings for the committee is dependent on how much risk the business is willing to take, and the number of AI projects in the organization. A proper governance process needs to be set up to reduce the risk of conflict of interest. If resources are available, external auditors should be considered, and announced if appropriate to build trust.
- **3.3.4 Stakeholder Engagement (3 mins):** Involving employees, customers, and partners in discussions about AI's impact. This could be Internal Town Halls, Dedicated Intranet Channels, Customer Advisory Board, Customer Feedback Channels, Shared Knowledge Base with partners. Building a culture of collaboration, and cross-functional team that accept feedback from all stakeholders, be able to deliberate and act on constructive feedback.
- **3.3.5 Continuous Monitoring & Audit (4 mins):** Regularly assessing AI systems for performance, fairness, and potential issues. Skills and knowledge will be needed to conduct well. How often should the audit be conducted, and how many resources are needed to conduct the audit well depends on the risk appetite of business. Again the audit processes needs to be set up properly with the necessary audit reminders, and reduce conflict of interest.

Module 4: Key Considerations & Framework for Successful AI Deployment

Suggested Duration: 49 mins

The final module equips you with a framework to anticipate and handle the broader challenges of AI adoption. We will explore the evolving legal and regulatory landscape and its potential impact on your business. The module addresses how AI changes the workplace, with a focus on upskilling your team and managing the change effectively. You'll also learn how to measure the true return on investment (ROI) of your AI projects, considering both short-term and long-term benefits. Finally, the module provides guidance on future-proofing your business by staying updated on AI trends and fostering an AI-ready culture.

Goal: Equip business owners with a framework to anticipate and mitigate broader risks associated with AI adoption.

Learning Objectives:

Upon completion of this module, business owners will be able to:

- Learn to anticipate and plan for challenges beyond technology, including the evolving legal landscape and a long-term vision for integrating AI into your core business strategy.
- Formulate strategies for effective change management, upskilling employees, and fostering an "AI-ready" organizational culture.
- Develop methods to measure both the tangible and intangible return on investment (ROI) of your AI initiatives.
- Identify and recognize the value of strategic partnerships to accelerate and optimize your AI adoption journey.
- Develop a mindset of continuous learning and adaptation to keep pace with emerging AI trends and maintain a competitive edge.

Module Outline:

- **4.1 Legal & Regulatory Landscape**
 - **4.1.1 Emerging Regulations (4 mins):** Awareness of evolving AI laws (e.g., EU AI Act, Singapore's Model AI Governance Framework).
 - **4.1.2 Industry Standards & Best Practices (3 mins):** Adhering to relevant guidelines for responsible AI.

- **4.1.3 Compliance & Audit Readiness (5 mins):** Preparing for potential regulatory scrutiny. Documentation is important and provides some guidelines on doing good documentation.
- **4.2 Human-AI Collaboration & Workforce Impact**
 - **4.2.1 Upskilling & Reskilling (4 mins):** Training employees to work alongside AI and adapt to new roles. State that the curation of the training content and trainer's background will be important.
 - **4.2.2 Shifting Job Roles (4 mins):** How AI can change existing job descriptions and create new opportunities. This needs employees to keep abreast of the latest changes in AI.
 - **4.2.3 Change Management Strategies (5 mins):** Effectively communicating AI adoption to employees and addressing concerns. Start with addressing the uncertainty of jobs possibly faced by the employees, moving to the detailed change transition to manage staff expectations.
- **4.3 Measuring Return on Investment (ROI) of AI**
 - **4.3.1 Quantifying Value (5 mins):** Measuring both tangible (cost savings, revenue increase) and intangible benefits (customer satisfaction, employee morale). Good quality data is needed to enable measurement.
 - **4.3.2 Long-term vs. Short-term Gains (3 mins):** Understanding that some AI benefits may take time to materialize.
 - **4.3.3 Continuous Evaluation (5 mins):** Regularly assessing AI performance against business goals. Benchmarking will be important.
- **4.4 Future Proofing Your Business with AI**
 - **4.4.1 Staying Updated on AI Trends (4 mins):** Importance of continuous learning and monitoring new developments.
 - **4.4.2 Fostering an AI-Ready Culture (3 mins):** Promoting innovation and embracing technological change. Important to increase the AI literacy level of the whole business organization to continuously extract value.
 - **4.4.3 Strategic Partnerships (4 mins):** Collaborating with AI experts and technology providers. How to select the good ones?

Optional Modules

Module 5: Advanced Data Management & Governance for AI

Suggested Duration: 67 mins

This module provides a deeper dive into the importance of high-quality data for AI success. It is crucial for businesses that want to move beyond basic AI applications and build more sophisticated, data-driven solutions.

Goal: Provide a deeper dive into the practical aspects of managing and governing data to optimize AI performance and ensure compliance.

Learning Objectives:

Upon completion of this module, business owners will be able to:

- Describe the full data lifecycle in the context of AI (acquisition, storage, transformation, versioning) and identify key considerations at each stage for optimal AI performance.
- Develop a more robust understanding of data governance principles (ownership, quality, security, compliance) and their direct impact on AI project success and ethical adherence.
- Evaluate common data management challenges (e.g., data silos, quality issues) and propose strategic solutions relevant to their business.
- Recognize the critical importance of specific data privacy regulations like Singapore's PDPA and, if applicable, GDPR, in managing data for AI initiatives.

Module Outline:

- **5.1 Data Lifecycle Management for AI:**
 - **5.1.1 Data Acquisition & Ingestion (8 mins):** Strategies for collecting relevant data from various sources (internal, external, third-party APIs).
 - **5.1.2 Data Storage & Infrastructure (8 mins):** Overview of scalable storage solutions (cloud data lakes, data warehouses) suitable for AI workloads.
 - **5.1.3 Data Transformation & Preparation (8 mins):** Techniques for cleaning, normalizing, and preparing data for AI model training (e.g., handling missing values, outliers).
 - **5.1.4 Data Versioning & Lineage (5 mins):** Importance of tracking data changes and understanding data origin for reproducibility and auditing.

- **5.2 Robust Data Governance Frameworks for AI:**
 - **5.2.1 Data Ownership & Stewardship (5 mins):** Clearly defining roles and responsibilities for data assets.
 - **5.2.2 Data Quality Management (5 mins):** Implementing processes and tools to ensure data accuracy, consistency, and completeness.
 - **5.2.3 Data Security & Access Control (5 mins):** Best practices for protecting sensitive data from unauthorized access and cyber threats.
 - **5.2.4 Compliance & Regulatory Landscape (Deep Dive) (8 mins):** Understanding specific data privacy laws (e.g., PDPA in Singapore, GDPR if applicable internationally) and their implications for AI data.
 - **5.2.5 Data Archiving & Retention Policies (5 mins):** Managing data throughout its lifecycle for compliance and efficiency.
- **5.3 Tools and Technologies for Data Management:**
 - **5.3.1 Overview of Data Platforms (5 mins):** Introduction to common cloud data services (e.g., AWS S3/Glue, Azure Data Lake/Synapse, Google Cloud Storage/BigQuery).
 - **5.3.2 Data Governance Tools (5 mins):** Brief mention of platforms that aid in data cataloging, quality monitoring, and access management.

Module 6: Building an AI-Ready Culture & Workforce Transformation

Suggested Duration: 56 mins

This module addresses the human side of AI adoption. It is important for business owners to learn how to prepare their team for AI implementation, manage change effectively, and create a culture that embraces new technology.

Goal: Focus on the human element of AI adoption, emphasizing cultural shifts, skill development, and effective change management.

Learning Objectives:

Upon completion of this module, business owners will be able to:

- Formulate strategies for cultivating an organizational culture that embraces AI, encourages experimentation, and fosters continuous learning.
- Analyze the potential impact of AI on their workforce (job augmentation, new roles) and develop a preliminary plan for upskilling and reskilling employees.
- Implement effective change management techniques to ensure smooth AI adoption, address employee concerns, and minimize resistance within the organization.
- Understand the critical role of human-AI collaboration and maintaining human oversight in maximizing the benefits and ethical deployment of AI.

Module Outline:

- **6.1 Cultivating an AI-First Mindset:**
 - **6.1.1 Leadership Buy-in & Vision (5 mins):** How leaders can champion AI and communicate its strategic importance. Leadership is important to establish culture and increase momentum in AI adoption.
 - **6.1.2 Promoting Experimentation & Learning (5 mins):** Encouraging a safe environment for AI pilots and continuous improvement. This is especially important since using AI is all about iterative learning and experimentation.
 - **6.1.3 Data Literacy Across the Organization (5 mins):** Empowering all employees with basic data understanding and interpretation skills, to know how data works which is the precursor to AI literacy.
- **6.2 Workforce Planning & Skill Development:**
 - **6.2.1 Identifying AI's Impact on Roles (6 mins):** Analyzing how AI will augment, change, or create new job functions within the business. How will LLMs or GenAI now and later impact the job paradigm?

- **6.2.2 Upskilling & Reskilling Programs (5 mins):** Strategies for training current employees in AI-related skills (e.g., data analysis, AI tool usage, prompt engineering). After training in technical skills, is to improve on soft skills that cannot be replaced by AI easily.
- **6.2.3 Attracting & Retaining AI Talent (5 mins):** Understanding the market for AI professionals and strategies for hiring. This is the first step, followed by retaining talents. This is especially important going forward where good talents are rare.
- **6.2.4 Human-in-the-Loop Design (6 mins):** Emphasizing the continuous need for human oversight and collaboration with AI systems. This is the last line against GenAI hallucination.
- **6.3 Effective Change Management for AI Adoption:**
 - **6.3.1 Communication Strategies (5 mins):** Clearly articulating the "why" and "how" of AI adoption to mitigate fear and build excitement. Manage the uncertainty then followed by what are the benefits to look out for.
 - **6.3.2 Addressing Employee Concerns (5 mins):** Proactively tackling fears about job displacement and ensuring a just transition.
 - **6.3.3 Creating AI Champions (4 mins):** Identifying and empowering internal advocates for AI within the organization. Showcase those who have benefitted much from the organization's adoption of AI.
 - **6.3.4 Feedback Mechanisms (5 mins):** Establishing channels for employees to provide input and contribute to AI development. Show that the feedback matters and acted upon.

Module 7: AI Project Risk Management & Mitigation

Suggested Duration: 75 mins

This module offers an advanced look at identifying and managing the specific risks associated with AI projects. It is essential for business owners who want to minimize potential pitfalls and ensure their AI investments are secure and successful.

Goal: Equip business owners with the knowledge and tools to identify, assess, and proactively mitigate common risks throughout the AI project lifecycle.

Learning Objectives:

Upon completion of this module, business owners will be able to:

- **Categorize and explain** the various types of risks specific to AI projects, including technical, operational, financial, ethical, reputational, and security risks.
- **Apply basic methods** for identifying, assessing (likelihood and impact), and prioritizing potential AI project risks within their specific business context.
- **Identify and propose practical strategies** to mitigate common AI risks across the entire AI project lifecycle, from initial planning to post-deployment.
- **Understand the importance** of proactive planning, iterative development, robust data quality controls, human oversight, and continuous monitoring in reducing AI project risks.
- **Recognize the role of Explainable AI (XAI)** and strong cybersecurity practices in enhancing risk mitigation efforts for AI systems.

Module Outline:

- **7.1 Understanding AI-Specific Risks:**
 - **7.1.1 Technical Risks (8 mins):** Model accuracy issues, scalability challenges, integration complexities with existing systems, data drift (model performance degradation over time).
 - **7.1.2 Operational Risks (5 mins):** Unexpected workflow disruptions, maintenance overheads, system downtime, lack of clear ownership for AI outputs.
 - **7.1.3 Financial Risks (5 mins):** Underestimated costs (data prep, infrastructure, ongoing maintenance), lower-than-expected ROI, failure to secure funding.
 - **7.1.4 Ethical & Reputational Risks (Deeper Dive) (5 mins):** Beyond basic awareness, understanding the *mechanisms* of bias, privacy breaches, and lack of transparency leading to real-world harm or public backlash.

- **7.1.5 Security Risks (5 mins):** Vulnerabilities in AI models, adversarial attacks (manipulating AI inputs), data poisoning, and risks to intellectual property within AI systems.
- **7.2 AI Risk Assessment & Prioritization:**
 - **7.2.1 Identifying Potential Pitfalls (6 mins):** Techniques for brainstorming and documenting risks at each phase of an AI project (from ideation to deployment).
 - **7.2.2 Quantifying Risk (4 mins):** Simple methods for assessing likelihood and impact of identified risks. Use past data or similar lessons wherever possible, depending on the maturity of the documentation process.
 - **7.2.3 Prioritizing Risks (3 mins):** Focusing on high-impact, high-likelihood risks first.
- **7.3 Strategies for AI Risk Mitigation:**
 - **7.3.1 Proactive Planning & Due Diligence (5 mins):** Thorough vendor evaluation, realistic goal setting, and comprehensive resource allocation.
 - **7.3.2 Robust Data Governance & Quality Control (4 mins):** Implementing measures to ensure data integrity and prevent bias (e.g., data audits, diverse datasets).
 - **7.3.3 Iterative Development & Testing: (4 mins)** Agile approaches, continuous testing (including adversarial testing), and pilot programs to identify issues early.
 - **7.3.4 Explainable AI (XAI) Adoption (5 mins):** Where relevant, choosing or designing AI solutions that provide transparency into their decision-making process. Document the lessons learnt, or the decision made during the AI project execution.
 - **7.3.5 Human Oversight & Intervention (3 mins):** Designing systems that allow for human review, override, and intervention, especially for critical decisions.
 - **7.3.6 Continuous Monitoring & Post-Deployment Audits (3 mins):** Setting up systems to track AI model performance, detect anomalies, detect bias, and identify new risks post-launch.
 - **7.3.5 Cybersecurity Best Practices for AI (4 mins):** Securing AI infrastructure, models, and data against cyber threats. Make sure the prompts do not contain any sensitive information of individuals and organizations.
- **7.4 Learning from Failure & Adapting:**
 - **7.4.1 Post-Mortem Analysis (3 mins):** Conducting reviews of AI projects that didn't meet expectations to identify lessons learned.
 - **7.4.2 Agility & Flexibility (3 mins):** Building an organizational capacity to adapt AI strategies based on new information and evolving risks.

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We would like to acknowledge the valuable support and collaboration from the AVPN, Google.org, ADB and all other partners who made this curriculum possible.

Your feedback is important to us! Please share your thoughts and suggestions to help us improve the curriculum. Reach us at aimaseanfoundation@gmail.com

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